

Final Model Report: NYC Uber/Lyft Hourly Demand Forecasting

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Abstract

This report presents the final demand forecasting model for hourly Uber and Lyft demand across New York City pickup zones. The final framework combines a primary XGBoost demand forecasting model with a residual correction model designed specifically for high-demand observations.

The final model achieves an overall mean absolute error (MAE) of 8.4533, root mean squared error (RMSE) of 15.9376, and coefficient of determination (R^2) of 0.9516. Model performance remains particularly strong for observations between 50 and 500 trips per zone-company-hour. However, extreme demand above 500 trips remains the primary source of forecasting error, with a normalized MAE of 12% and an underprediction rate of 53%.

The results suggest that the model captures the majority of regular temporal demand patterns effectively, while extreme demand events remain more difficult to predict because of their relatively low frequency and larger deviations from historical demand patterns.

1 Whole Demand Level Analysis

The whole demand levels are separated from 0 to 5, 5 to 10, 10 to 50, and so on. When the demand goes up, the magnitude of the error (namely, NMAE) goes down. The prediction is relatively accurate at mid-level demand, and the high-demand situation is finally improved by an additional correction model for high-demand situation.

Table 1: Prediction Error by Demand Level

Demand Level	Observations	Avg. Demand	Avg. Predicted Demand	Avg. Residual	MAE	Total Demand	Normalized MAE
0-5	79,204	2.78	4.19	-1.41	1.95	219,814	0.703
5-10	61,556	7.95	9.25	-1.30	2.97	489,446	0.374
10-50	305,179	27.38	29.03	-1.65	5.84	8,354,801	0.213
50-100	149,991	71.50	72.00	-0.49	9.20	10,724,828	0.129
100-200	91,639	139.02	135.48	3.53	15.43	12,739,323	0.111
200-500	30,531	273.33	260.46	12.87	33.54	8,345,091	0.123
500+	1,961	634.99	595.66	39.34	76.27	1,245,225	0.120

2 Correction Model Training Data

The high-demand validation sample contains 2,879 observations.

Table 2: High-Demand Correction Data

Statistic	Value
High-demand observations	2,879
Average actual demand	584.7100
Average main-model prediction	538.8819
Average correction target	45.8281

The positive average correction target indicates that the main model systematically underestimated demand within the high-demand correction sample. On average, actual demand exceeded the main-model prediction by approximately 45.83 trips.

3 Overall Final Model Performance

The final model produces the following out-of-sample performance:

Table 3: Final Model Performance

Metric	Value
MAE	8.4563
RMSE	15.7416
R^2	0.9516

The MAE indicates that the model’s prediction differs from observed hourly demand by approximately 8.45 trips on average.

The RMSE is larger than the MAE because large forecasting errors receive greater penalties under the squared-error formulation. This difference indicates that a relatively small number of high-error observations remain in the validation data.

The R^2 of 0.9504 indicates that approximately 95.04% of the variation in demand within the evaluation sample is explained by the model.

4 Forecasting Error by Demand Level

Model performance varies considerably across demand regimes.

Define normalized MAE as

$$\text{NMAE} = \frac{\sum_{i=1}^n |D_i - \hat{D}_i|}{\sum_{i=1}^n D_i}. \quad (1)$$

4.1 Interpretation

The model performs particularly well in the medium- and high-demand ranges between 50 and 500 trips. Normalized MAE is approximately 11–13% across these observations.

The very high normalized error for the 0–5 demand category should be interpreted carefully. Because the denominator is extremely small, even an absolute error of only one or two trips can produce a large normalized percentage error. The absolute MAE in this category remains only

Table 4: Prediction Direction by Demand Level

Demand Level	Overprediction	Underprediction
0–5	0.729496	0.270504
5–10	0.583647	0.416353
10–50	0.551257	0.448743
50–100	0.496336	0.503664
100–200	0.407970	0.592030
200–500	0.330779	0.669221
500+	0.374809	0.625191

1.95 trips. In addition to this, small demand does not have a large business impact because the demand level is relatively small and it is less lucrative.

Extreme demand above 500 trips presents an improvement when I used the additional correction model using XGboost. Average actual demand is approximately 634 trips, while the average prediction is approximately 595 trips. Therefore, the remaining error is somehow acceptable but there can be a room for advanced improvement.

5 Overprediction and Underprediction

The direction of forecasting error provides additional information beyond absolute-error metrics.

Table 5: Prediction Direction by Demand Level

Demand Level	Overprediction	Underprediction
0–5	72.95%	27.05%
5–10	58.36%	41.64%
10–50	55.13%	44.87%
50–100	49.63%	50.37%
100–200	40.80%	59.20%
200–500	33.34%	66.92%
500+	37.48%	62.51%

A clear directional pattern is observed. The model tends to overpredict very low demand and increasingly underpredict demand as the actual demand level rises.

After the correction model is applied, For demand above 500 trips, approximately 53.39% of observations are underpredicted. This indicates that extreme demand remains the primary regime in which systematic prediction bias persists.

6 Extreme Demand Analysis

For observations with actual demand exceeding 500 trips:

Table 6: Performance for Demand Above 450 Trips

Statistic	Value
Observations	2879
Average Actual Demand	584.7
Average Prediction	538.88
MAE	75.01
NMAE	12.8274%
Underprediction Rate	53.58%

Despite the residual correction mechanism, extreme demand remains more difficult to forecast than regular demand. Nevertheless, an NMAE of 16.49% indicates that the predictions retain substantial information about the magnitude of these high-demand periods.

7 Spatial Analysis of Extreme Demand

Extreme-demand observations are concentrated in a relatively small number of major transportation and commercial zones.

Table 7: High-Demand Prediction Performance by Pickup Zone

Pickup Zone	Observations	Avg. Demand	Avg. Prediction	Avg. Residual	MAE
LaGuardia Airport	597	667.74	629.06	38.68	92.18
JFK Airport	195	602.29	586.19	16.10	58.17
Midtown Center	139	701.17	692.96	8.20	49.80
East Village	134	719.54	682.65	36.88	82.65
Lower East Side	101	684.72	647.27	37.45	86.55
TriBeCa/Civic Center	82	567.04	507.94	59.09	63.99
Union Sq	72	554.67	529.87	24.79	41.92
West Chelsea/Hudson Yards	70	587.94	538.47	49.47	59.00
Williamsburg (North Side)	69	595.99	561.85	34.13	60.62
Crown Heights North	69	578.29	528.25	50.04	62.32
Times Sq/Theatre District	64	584.36	586.35	-1.99	46.60
Bushwick South	62	654.94	603.80	51.13	83.33
West Village	53	593.49	566.07	27.42	47.54
Greenpoint	47	571.04	541.33	29.71	63.17
Midtown East	41	609.83	579.58	30.25	46.98
Bushwick North	35	621.49	548.25	73.23	97.36

LaGuardia Airport contains the largest number of extreme-demand observations, followed by JFK Airport. Extreme demand is also concentrated in major Manhattan commercial and nightlife areas, including Midtown Center, East Village, Lower East Side, and West Chelsea/Hudson Yards.

The spatial results also reveal substantial heterogeneity in forecasting difficulty. For example, average residuals are considerably larger in several nightlife and residential-commercial zones than at JFK Airport. This suggests that extreme demand may be generated by different mechanisms across zones.

8 Holiday Performance

Table 8: Holiday and Non-Holiday Forecast Performance

Holiday	Observations	Avg. Demand	Avg. Prediction	MAE
No	684,774	58.912	58.629	8.384
Yes	35,287	50.361	50.039	9.790

Average predictions remain close to average observed demand for both holiday and non-holiday periods. However, holiday observations have a higher MAE of 9.79 compared with 8.38 for non-holiday observations.

This indicates that holiday demand contains greater observation-level variation even though the model remains approximately unbiased at the aggregate level.

9 Feature Importance

The XGBoost feature importance results are shown below.

Table 9: Final Model Feature Importance

Feature	Importance
Lag 168 Hours	0.4357
Lag 1 Hour	0.3974
Lag 24 Hours	0.0770
Rolling Mean 3 Hours	0.0364
Zone-Hour Interaction	0.0147
Hour	0.0109
Day of Week	0.0036
Pickup Zone	0.0032
Rolling Maximum 24 Hours	0.0032
Holiday Indicator	0.0028
Rolling Mean 24 Hours	0.0025
Weekend Indicator	0.0020
Rain Category	0.0018
Rolling Mean 168 Hours	0.0016
Temperature	0.0016
Snow Indicator	0.0016
Month	0.0015
Wind Category	0.0014
Company	0.0012

9.1 Interpretation of Feature Importance

Historical demand dominates the predictive structure of the model. The two most important predictors are demand from the same hour one week earlier and demand from the immediately preceding hour.

Together,

$$0.4357 + 0.3974 \approx 0.8331, \quad (2)$$

meaning that these two variables account for approximately 83.3% of the reported XGBoost feature importance.

The importance of Lag168 demonstrates strong weekly periodicity in ride demand, while Lag1 captures short-term demand persistence. The importance of Lag24 provides additional evidence of daily seasonality.

Weather variables have relatively low individual feature importance. This does not necessarily imply that weather has no effect on ride demand. Instead, much of its predictive contribution may be smaller than, conditional on, or partially captured by recent historical demand patterns.

10 Business Interpretation

The model produces three findings that are particularly relevant for operational demand forecasting.

1. Regular demand is highly predictable.

For demand between 50 and 500 trips per zone-company-hour, normalized MAE remains approximately 11–13%. This suggests that historical demand patterns provide a strong operational baseline for forecasting normal and moderately high demand.

2. Extreme demand remains systematically underestimated.

For demand above 500 trips, the model underpredicts approximately 81.39% of observations. Therefore, an operational system relying only on point forecasts should account for additional uncertainty during extreme-demand periods.

3. Extreme-demand errors are spatially concentrated.

Airports, major Manhattan commercial areas, and several nightlife zones account for a substantial portion of observations above 500 trips. Consequently, further model improvements could focus on zone-specific extreme-demand behavior rather than applying a uniform correction across all New York City zones.

11 Model Limitations

Several limitations remain in the final forecasting framework.

First, extreme-demand observations are relatively rare compared with normal-demand observations. Only 2848 validation observations exceed 500 trips, making this regime substantially more difficult for the model to learn.

Second, the current feature set relies heavily on historical demand. Unexpected events such as concerts, major sporting events, flight disruptions, transit interruptions, or unusual local activity may generate demand shocks that cannot be fully inferred from lagged demand.

Third, standard XGBoost feature importance measures predictive usage within the fitted model and should not be interpreted as causal effects. In particular, the relatively low importance assigned to weather variables does not demonstrate that weather has no causal effect on ride demand.

Finally, the residual correction model reduces systematic high-demand bias but does not eliminate it. The remaining 53.58% underprediction rate for demand above 500 indicates that extreme-demand forecasting remains the primary opportunity for future improvement.

12 Potential Future Improvements

Future extensions could incorporate additional predictors related to extreme-demand events, including

- major concerts and sporting events,
- airport flight arrival and departure volume,
- public transportation disruptions,
- major holidays and special events,
- zone-specific high-demand correction models,
- interaction effects between location, hour, and weather, and
- probabilistic prediction intervals for operational uncertainty.

A particularly useful extension would be to model extreme demand separately for airports, Manhattan commercial zones, and nightlife areas. The spatial analysis suggests that the mechanisms generating extreme demand differ substantially across these locations.

13 Final Conclusion

The final model demonstrates strong overall forecasting performance, but systematic underprediction remains during extreme-demand periods. In particular, demand above 500 trips is underpredicted in approximately 53.59% of observations. However, these extreme-demand observations are not distributed uniformly across New York City. They are concentrated in several high-volume locations, including LaGuardia Airport, JFK Airport, Midtown, East Village, and other major commercial or nightlife zones.

Rather than uniformly increasing predictions across all zones, a more practical operational strategy is to introduce a **targeted demand buffer** for locations with historically persistent high-demand underprediction. For a selected set of high-risk zones $\mathcal{Z}_H$, an operational forecast can be expressed as

$$\hat{D}_{z,c,t}^{\text{operational}} = \hat{D}_{z,c,t}^{\text{final}} + I(z \in \mathcal{Z}_H)B_{z,t}, \quad (3)$$

where $B_{z,t}$ represents an additional demand buffer determined from historical residual patterns for zone z and potentially varying by time of day.

This approach provides additional capacity protection in locations where extreme demand is repeatedly underestimated without unnecessarily increasing predictions throughout the entire system.

At the same time, the buffer should remain conservative. Excessive upward adjustment may create **overprediction**, which can result in too many drivers being directed toward locations where realized demand is insufficient. This can increase driver idle time, empty vehicle mileage, congestion, and inefficient allocation of available supply. Therefore, completely eliminating underprediction should not necessarily be the operational objective.

A more appropriate objective is to balance the asymmetric costs of forecasting errors. Moderate underprediction can often be addressed through real-time driver repositioning and supply responses,

while substantial overprediction may proactively allocate resources to demand that never materializes. Consequently, the recommended strategy is to preserve the current model as the primary forecast while applying **small, zone-specific buffers only to locations and periods with persistent extreme-demand underprediction.**

Overall, the model should therefore be viewed not only as a point forecasting system, but as a decision-support framework in which statistical predictions are combined with targeted operational adjustments. This approach reduces exposure to extreme-demand shortages while limiting the operational cost associated with unnecessary overprediction.